

Offline HTML Documentation (Reference)

1. Introduction

This document is an offline, compact HTML reference for paragliding instructors. It provides quick access to HTML element definitions, attributes, and related concepts without requiring an internet connection.

This document is NOT a tutorial or learning guide. It assumes a basic understanding of HTML and focuses on providing reference information.

Standards Referenced: * HTML Living Standard (WHATWG) * HTML5

2. HTML Document Structure

Required Elements

Every HTML document must contain: * `<!DOCTYPE html>` : The document type declaration. * `<html>` : The root element. * `<head>` : Contains meta-information about the document. * `<body>` : Contains the visible document content.

Optional Elements

Elements within `<head>` are generally optional but crucial for functionality and metadata.

Doctype

The `<!DOCTYPE html>` declaration is mandatory. It tells the browser that the document is an HTML5 document.

DOM Overview

The Document Object Model (DOM) is a programming interface for HTML and XML documents. It represents the page so that programs can change the document structure, style, and content. It's a tree-like structure where each node represents a part of the document (e.g., elements, text).

3. Global Attributes

These attributes can be applied to most HTML elements.

- `accesskey`: `[a-z0-9]` - Assigns a keyboard shortcut to an element.
- `autocapitalize`: `off`, `none`, `sentences`, `words`, `characters` - Controls automatic capitalization.
- `autofocus`: (boolean) - Automatically focuses an element when the page loads.
- `class`: `<string>` - A space-separated list of classes, used for CSS styling and JavaScript selection.
- `contenteditable`: `true`, `false`, `plaintext-only` - Makes the element editable by the user.
- `data-*`: `<string>` - Custom data attributes, used to store private data related to the element.
- `dir`: `ltr`, `rtl` - Sets the text direction of the element.
- `draggable`: `true`, `false` - Indicates if the element can be dragged.
- `enterkeyhint`: `enter`, `done`, `go`, `next`, `previous`, `search`, `send` - Suggests the label for the Enter key on virtual keyboards.
- `exportparts`: `<string>` - Specifies parts of a shadow DOM to be exported.
- `hidden`: (boolean) - Hides the element.
- `id`: `<string>` - A unique identifier for the element within the document.
- `inert`: (boolean) - Makes the element and its descendants non-interactive and unsearchable.
- `inputmode`: `none`, `text`, `tel`, `url`, `email`, `numeric`, `decimal`, `search` - Suggests the type of keyboard to display for form inputs.

- `is: <string>` - Used for custom elements, specifying which custom element a standard HTML element should be treated as.
- `itemid, itemprop, itemscope, itemtype`: (Schema.org related) - Used for microdata.
- `lang: <language-code>` - Specifies the language of the element's content.
- `nonce: <string>` - A cryptographic nonce for script and style execution.
- `part: <string>` - Names parts of a shadow DOM for styling.
- `popover`: (boolean, or `manual` , `auto`) - Makes an element a popover.
- `role: <string>` - Defines the role of an element for accessibility.
- `slot: <string>` - Assigns a slot to an element within a shadow DOM.
- `spellcheck: true , false` - Enables or disables spell checking.
- `style: <string>` - Inline CSS styles.
- `tabindex: <integer>` - Determines the tab order of the element.
- `title: <string>` - Provides advisory information about the element, often displayed as a tooltip.
- `translate: yes , no` - Specifies whether an element's attribute values and content should be translated.

4. Core Element Reference

4.1 Document Metadata

- **`<html>`**
 - Categories: Root element.
 - Permitted content: `<head>` , `<body>` .
 - Important attributes: `lang` , `dir` .
 - Common use cases: The top-level container for all HTML content.
- **`<head>`**
 - Categories: Metadata content.
 - Permitted content: `<base>` , `<link>` , `<meta>` , `<style>` , `<script>` , `<noscript>` , `<title>` , `<template>` .
 - Important attributes: None.

- Common use cases: Defines metadata for the document.

- **<title>**

- Categories: Metadata content, flow content.
- Permitted content: Phrasing content.
- Important attributes: None.
- Common use cases: Sets the title shown in the browser tab or window.

- **<base>**

- Categories: Metadata content.
- Permitted content: None.
- Important attributes: href , target .
- Common use cases: Specifies a base URL for all relative URLs in a document.

- **<link>**

- Categories: Metadata content.
- Permitted content: None.
- Important attributes: href , rel , type , media .
- Common use cases: Links to external resources like stylesheets (rel="stylesheet").

- **<meta>**

- Categories: Metadata content.
- Permitted content: None.
- Important attributes: charset , name , content , http-equiv .
- Common use cases: Specifies character set (charset="UTF-8"), viewport settings, description, keywords.

- **<style>**

- Categories: Metadata content, scripting content.
- Permitted content: Phrasing content (ignored).
- Important attributes: media , nonce .

- Common use cases: Embeds CSS styles directly in the document.
- **<script>**
 - Categories: Metadata content, scripting content.
 - Permitted content: Phrasing content (ignored).
 - Important attributes: `src` , `async` , `defer` , `type` , `nonce` .
 - Common use cases: Embeds or links to JavaScript code.
- **<noscript>**
 - Categories: Metadata content, scripting content.
 - Permitted content: Flow content.
 - Important attributes: None.
 - Common use cases: Provides fallback content for users with JavaScript disabled.
- **<template>**
 - Categories: Metadata content, scripting content.
 - Permitted content: Flow content.
 - Important attributes: None.
 - Common use cases: Holds client-side content that is not to be rendered when a page is loaded but can be instantiated later.

4.2 Sections

- **<body>**
 - Categories: Metadata content, flow content.
 - Permitted content: Flow content.
 - Important attributes: `onafterprint` , `onbeforeprint` , `onbeforeunload` , `onhashchange` , `onlanguagechange` , `onmessage` , `onpagehide` , `onpageshow` , `onpopstate` , `onstorage` , `onunload` .
 - Common use cases: Contains all visible content of an HTML document.

- **<article>**

- Categories: Sectioning content, flow content.
- Permitted content: Flow content.
- Important attributes: None.
- Common use cases: Represents a self-contained composition in a document, page, application, or site, intended to be independently distributable or reusable.

- **<section>**

- Categories: Sectioning content, flow content.
- Permitted content: Flow content.
- Important attributes: None.
- Common use cases: Represents a standalone section of a document, typically with a heading.

- **<nav>**

- Categories: Sectioning content, flow content.
- Permitted content: Flow content.
- Important attributes: None.
- Common use cases: Represents a section of a page that links to other pages or to parts within the page.

- **<aside>**

- Categories: Sectioning content, flow content.
- Permitted content: Flow content.
- Important attributes: None.
- Common use cases: Represents content that is tangentially related to the content around it, often presented as a sidebar or pull-quote.

- **<h1> – <h6>**

- Categories: Sectioning content, flow content.
- Permitted content: Phrasing content.
- Important attributes: None.
- Common use cases: Define document headings, with **<h1>** being the most important and **<h6>** the least.

- **<header>**
 - Categories: Sectioning content, flow content.
 - Permitted content: Flow content.
 - Important attributes: None.
 - Common use cases: Represents introductory content, typically a group of introductory or navigational aids.
- **<footer>**
 - Categories: Sectioning content, flow content.
 - Permitted content: Flow content.
 - Important attributes: None.
 - Common use cases: Represents a footer for its nearest sectioning content or root element.
- **<address>**
 - Categories: Sectioning content, flow content.
 - Permitted content: Phrasing content.
 - Important attributes: None.
 - Common use cases: Represents the contact information for its nearest **<article>** or **<html>** ancestor.
- **<main>**
 - Categories: Sectioning content, flow content.
 - Permitted content: Flow content.
 - Important attributes: None.
 - Common use cases: Represents the dominant content of the **<body>** of a document.

4.3 Grouping Content

- **<p>**
 - Categories: Flow content.
 - Permitted content: Phrasing content.
 - Important attributes: None.

- Common use cases: Represents a paragraph of text.

- **<hr>**

- Categories: Flow content.
- Permitted content: None.
- Important attributes: None.
- Common use cases: Represents a thematic break between paragraph-level content.

- **<pre>**

- Categories: Flow content.
- Permitted content: Phrasing content.
- Important attributes: None.
- Common use cases: Represents preformatted text, preserving whitespace and line breaks.

- **<blockquote>**

- Categories: Flow content.
- Permitted content: Flow content.
- Important attributes: `cite` .
- Common use cases: Represents content quoted from another source.

- ****

- Categories: Flow content.
- Permitted content: `<li>` , `<script>` , `<template>` .
- Important attributes: `type` , `reversed` , `start` .
- Common use cases: Represents an ordered list.

- ****

- Categories: Flow content.
- Permitted content: `<li>` , `<script>` , `<template>` .
- Important attributes: None.
- Common use cases: Represents an unordered list.

- ****
 - Categories: Flow content.
 - Permitted content: Flow content.
 - Important attributes: `value` .
 - Common use cases: Represents a list item within `<ol>` or `<ul>` .
- **<dl> , <dt> , <dd>**
 - Categories: Flow content.
 - Permitted content: `<dl>` : `<dt>` , `<dd>` , `<script>` , `<template>` . `<dt>` : Phrasing content. `<dd>` : Flow content.
 - Important attributes: None.
 - Common use cases: Represents a description list. `<dt>` defines the term, `<dd>` defines the description.
- **<figure> , <figcaption>**
 - Categories: Flow content.
 - Permitted content: `<figure>` : Flow content. `<figcaption>` : Flow content.
 - Important attributes: None.
 - Common use cases: `<figure>` is used for self-contained content like images, diagrams, code listings, etc., that are referenced in the main text. `<figcaption>` provides a caption for the figure.
- **<div>**
 - Categories: Flow content.
 - Permitted content: Flow content.
 - Important attributes: None.
 - Common use cases: A generic container for flow content, used for grouping and styling.

4.4 Text-level Semantics

- **<a>**
 - Categories: Flow content, phrasing content.

- Permitted content: Phrasing content.
- Important attributes: href , target , rel , download .
- Common use cases: Creates hyperlinks to other pages or resources.

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- Categories: Phrasing content.
- Permitted content: Phrasing content.
- Important attributes: None.
- Common use cases: Emphasizes text, typically rendered in italics.

- ****

- Categories: Phrasing content.
- Permitted content: Phrasing content.
- Important attributes: None.
- Common use cases: Indicates strong importance, seriousness, or urgency, typically rendered in bold.

- **<small>**

- Categories: Phrasing content.
- Permitted content: Phrasing content.
- Important attributes: None.
- Common use cases: Represents side comments or small print.

- **<s>**

- Categories: Phrasing content.
- Permitted content: Phrasing content.
- Important attributes: None.
- Common use cases: Represents content that is no longer accurate or relevant.

- **<cite>**

- Categories: Phrasing content.
- Permitted content: Phrasing content.
- Important attributes: None.
- Common use cases: Represents the title of a creative work.

- **<q>**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.
 - Important attributes: `cite` .
 - Common use cases: Represents a short inline quotation.
- **<dfn>**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.
 - Important attributes: None.
 - Common use cases: Identifies the term being defined. The first occurrence of the term can be the definition.
- **<abbr>**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.
 - Important attributes: `title` .
 - Common use cases: Represents an abbreviation or acronym. The `title` attribute can provide the full expansion.
- **<ruby> , <rb> , <rt> , <rtc> , <rp>**
 - Categories: Phrasing content.
 - Permitted content: `<ruby>` : `<rb>` , `<rp>` , `<rt>` , `<rtc>` , `<template>` . `<rb>` , `<rt>` , `<rtc>` , `<rp>` : Phrasing content.
 - Important attributes: None.
 - Common use cases: Used for East Asian typography, enabling ruby annotations (short annotations rendered above or next to characters).
- **<data>**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.
 - Important attributes: `value` .
 - Common use cases: Links a piece of content with a machine-readable translation.

- **<time>**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.
 - Important attributes: `datetime`.
 - Common use cases: Represents a specific period in time or a date.
- **<code>**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.
 - Important attributes: None.
 - Common use cases: Represents a fragment of computer code.
- **<var>**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.
 - Important attributes: None.
 - Common use cases: Represents a variable in mathematics or a programming context.
- **<samp>**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.
 - Important attributes: None.
 - Common use cases: Represents sample output from a computer program.
- **<kbd>**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.
 - Important attributes: None.
 - Common use cases: Represents keyboard input.
- **<sub>** , **<sup>**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.

- Important attributes: None.
- Common use cases: Subscript (`<sub>`) and superscript (`<sup>`) text.

- **`<i>` , `<b>` , `<u>`**

- Categories: Phrasing content.
- Permitted content: Phrasing content.
- Important attributes: None.
- Common use cases: `<i>` : Typically rendered in italics, for stylistic reasons or foreign text. `<b>` : Typically rendered in bold, for stylistic reasons or to draw attention. `<u>` : Underlined text (use with caution to avoid confusion with links).

- **`<span>`**

- Categories: Phrasing content.
- Permitted content: Phrasing content.
- Important attributes: None.
- Common use cases: A generic inline container for phrasing content.

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- Categories: Phrasing content.
- Permitted content: None.
- Important attributes: None.
- Common use cases: Inserts a line break.

- **`<wbr>`**

- Categories: Phrasing content.
- Permitted content: None.
- Important attributes: None.
- Common use cases: Indicates a potential line break point, useful for long words that might otherwise overflow their container.

4.5 Edits

- **<ins>**
 - Categories: Flow content.
 - Permitted content: Flow content.
 - Important attributes: `cite` , `datetime` .
 - Common use cases: Represents text that has been inserted into the document.
- ****
 - Categories: Flow content.
 - Permitted content: Flow content.
 - Important attributes: `cite` , `datetime` .
 - Common use cases: Represents text that has been deleted from the document.

4.6 Embedded Content

- ****
 - Categories: Flow content, phrasing content.
 - Permitted content: None.
 - Important attributes: `src` , `alt` , `width` , `height` , `loading` .
 - Common use cases: Embeds an image. `alt` is crucial for accessibility.
- **<iframe>**
 - Categories: Flow content.
 - Permitted content: None.
 - Important attributes: `src` , `name` , `allow` , `sandbox` .
 - Common use cases: Embeds another document within the current one.
- **<embed>**
 - Categories: Flow content.

- Permitted content: None.
- Important attributes: `src` , `type` , `width` , `height` .
- Common use cases: Embeds external content, often plugins (though largely superseded by `<iframe>` and `<video>`).

- **`<object>`**

- Categories: Flow content.
- Permitted content: Phrasing content, but only `param` and fallback content.
- Important attributes: `data` , `type` , `width` , `height` .
- Common use cases: Embeds external resources, such as plugins or rich media.

- **`<video>`**

- Categories: Flow content.
- Permitted content: Phrasing content, `<source>` , `<track>` , `<script>` , `<template>` .
- Important attributes: `src` , `controls` , `width` , `height` , `autoplay` , `loop` , `muted` .
- Common use cases: Embeds video content.

- **`<audio>`**

- Categories: Flow content.
- Permitted content: Phrasing content, `<source>` , `<track>` , `<script>` , `<template>` .
- Important attributes: `src` , `controls` , `autoplay` , `loop` , `muted` .
- Common use cases: Embeds audio content.

- **`<track>`**

- Categories: Metadata content (within `<audio>` or `<video>`).
- Permitted content: None.
- Important attributes: `src` , `kind` , `srclang` , `label` , `default` .

- Common use cases: Provides text tracks for media elements (e.g., subtitles, captions).
- **<map>**
 - Categories: Flow content.
 - Permitted content: Flow content.
 - Important attributes: `name` .
 - Common use cases: Defines an image-map, used with the `<img>` `usemap` attribute.
- **<area>**
 - Categories: Phrasing content.
 - Permitted content: None.
 - Important attributes: `shape` , `coords` , `href` , `alt` .
 - Common use cases: Defines an area within an image-map.
- **<source>**
 - Categories: Phrasing content (within `<audio>` or `<video>`).
 - Permitted content: None.
 - Important attributes: `src` , `type` , `srcset` , `media` .
 - Common use cases: Provides multiple media resources for `<audio>` or `<video>` elements.

4.7 SVG and MathML (brief)

- **<svg>**
 - Categories: Embedded content.
 - Permitted content: SVG content.
 - Important attributes: `width` , `height` , `viewBox` .
 - Common use cases: Embeds Scalable Vector Graphics.
- **<math>**
 - Categories: Embedded content.
 - Permitted content: MathML content.

- Important attributes: None.
- Common use cases: Embeds mathematical formulas using MathML.

4.8 Table Content

- **<table>**

- Categories: Flow content.
- Permitted content: `<caption>` , `<colgroup>` , `<thead>` `(?, ?, ?)>` , `<tbody>` `(?, ?, ?)>` , `<tfoot>` `(?, ?, ?)>` , `<tr>` , `<script>` , `<template>` .
- Important attributes: `border` (deprecated), `summary` (deprecated).
- Common use cases: Defines a table.

- **<caption>**

- Categories: Table accessible content.
- Permitted content: Flow content.
- Important attributes: None.
- Common use cases: Provides a title or caption for a `<table>` .

- **<colgroup>** , **<col>**

- Categories: Table column group, table column.
- Permitted content: `<colgroup>` : `<col>` , `<script>` , `<template>` . `<col>` : None.
- Important attributes: `span` (for `<th>` and `<td>` if defined, but more commonly for `colgroup`), `style` .
- Common use cases: Groups columns in a table for styling or formatting.

- **<thead>** , **<tbody>** , **<tfoot>**

- Categories: Table section.
- Permitted content: `<tr>` , `<script>` , `<template>` .
- Important attributes: None.

- Common use cases: Organize table rows into header, body, and footer sections.
- **<tr>**
 - Categories: Table row.
 - Permitted content: `<th>` , `<td>` , `<script>` , `<template>` .
 - Important attributes: None.
 - Common use cases: Defines a row in a table.
- **<td>** , **<th>**
 - Categories: Table cell.
 - Permitted content: Flow content.
 - Important attributes: `colspan` , `rowspan` , `scope` (for `<th>`).
 - Common use cases: `<td>` : Table data cell. `<th>` : Table header cell.

4.9 Forms

- **<form>**
 - Categories: Flow content.
 - Permitted content: Phrasing content.
 - Important attributes: `action` , `method` , `enctype` , `accept-charset` , `autocomplete` , `novalidate` .
 - Common use cases: Creates an HTML form for user input.
- **<label>**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.
 - Important attributes: `for` , `form` .
 - Common use cases: Provides a label for a form control.
- **<input>**
 - Categories: Phrasing content, potentially interactive content.
 - Permitted content: None.

- Important attributes:

- `type` : `button` , `checkbox` , `color` , `date` , `datetime-local` , `email` , `file` , `hidden` , `image` , `month` , `number` , `password` , `radio` , `range` , `reset` , `search` , `submit` , `tel` , `text` , `time` , `url` , `week` .
- `value` : Default value for the input.
- `name` : Name of the input, sent with form submission.
- `list` : ID of a `<datalist>` element.
- `required` : Boolean, specifies if the input must be filled out.
- `minlength` , `maxlength` : Minimum/maximum length of text input.
- `pattern` : Regular expression for validating text input.
- `placeholder` : Hint for the expected value.
- `readonly` : Boolean, input is read-only.
- `disabled` : Boolean, input is disabled.
- `autofocus` : Boolean, input is automatically focused.
- `autocomplete` : Controls autocomplete behavior.
- `accept` : For `type="file"` , specifies file types.
- `multiple` : For `type="file"` , allows multiple files.
- `min` , `max` , `step` : For `type="number"` and `type="range"` .
- `rows` , `cols` : For `type="textarea"` .

- Common use cases: The most versatile form element, used for various input types.

- **`<button>`**

- Categories: Phrasing content, interactive content.
- Permitted content: Phrasing content.
- Important attributes: `type` (`submit` , `reset` , `button`), `name` , `value` , `disabled` .
- Common use cases: Creates a clickable button, often used to submit forms.

- **<select>**

- Categories: Phrasing content, potentially interactive content.
- Permitted content: `<optgroup>` , `<option>` , `<script>` , `<template>` .
- Important attributes: `name` , `size` , `multiple` , `required` , `autocomplete` .
- Common use cases: Creates a dropdown list of options.

- **<datalist>**

- Categories: Phrasing content.
- Permitted content: `<option>` , `<script>` , `<template>` .
- Important attributes: None.
- Common use cases: Provides a list of predefined options for an `<input>` element.

- **<optgroup>** , **<option>**

- Categories: Grouping options, option.
- Permitted content: `<optgroup>` : `<option>` , `<script>` , `<template>` . `<option>` : Phrasing content.
- Important attributes: `<optgroup>` : `label` . `<option>` : `value` , `label` , `selected` , `disabled` .
- Common use cases: Used within `<select>` to group or define options.

- **<textarea>**

- Categories: Phrasing content, potentially interactive content.
- Permitted content: Phrasing content.
- Important attributes: `name` , `rows` , `cols` , `placeholder` , `required` , `readonly` , `disabled` , `maxlength` , `autocomplete` .
- Common use cases: Creates a multi-line text input control.

- **<fieldset>** , **<legend>**

- Categories: Grouping content.

- Permitted content: `<fieldset>` : Flow content. `<legend>` : Phrasing content.
- Important attributes: `<fieldset>` : `name` , `form` , `disabled` .
- Common use cases: Used to group related form controls.
`<legend>` provides a caption for the fieldset.
- **`<output>`**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.
 - Important attributes: `name` , `form` , `for` .
 - Common use cases: Represents the result of a calculation or user action.
- **`<meter>`**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.
 - Important attributes: `value` , `min` , `max` , `low` , `high` , `optimum` .
 - Common use cases: Represents a scalar measurement within a known range (e.g., disk usage).
- **`<progress>`**
 - Categories: Phrasing content.
 - Permitted content: Phrasing content.
 - Important attributes: `value` , `max` .
 - Common use cases: Represents the progress of a task (e.g., upload progress).

4.10 Interactive Elements

- **`<details>`**
 - Categories: Flow content, interactive content.
 - Permitted content: `<summary>` , flow content.
 - Important attributes: `open` .

- Common use cases: Creates a disclosure widget where content can be revealed or hidden by the user.
- **<summary>**
 - Categories: Phrasing content (within `<details>`).
 - Permitted content: Phrasing content.
 - Important attributes: None.
 - Common use cases: Provides a caption or summary for a `<details>` element.
- **<dialog>**
 - Categories: Flow content, interactive content.
 - Permitted content: Flow content.
 - Important attributes: `open` .
 - Common use cases: Creates a dialog box, alert, or other interactive panel.
- **<menu>** , **<menuitem>**
 - Categories: Flow content.
 - Permitted content: `<menu>` : `menuitem` , `script` , `template` .
`<menuitem>` : None.
 - Important attributes: `<menu>` : `type` (`context` , `toolbar`), `label` . `<menuitem>` : `label` , `type` (`command` , `radio` , `checkbox`), `checked` , `disabled` , `icon` .
 - Common use cases: Legacy: Used to define menus. Modern usage is complex and often replaced by JavaScript-based solutions or new APIs.
- Popover API Notes
 - The `popover` attribute (`popover="manual"` or `popover="auto"`) along with the `:backdrop` pseudo-element and JavaScript API allows for the creation of temporary UI surfaces (like tooltips, menus, dialogs) that are managed by the browser.

4.11 Scripting

- **<script>** : (See 4.1 Document Metadata)
- **<canvas>**
 - Categories: Embedded content.
 - Permitted content: Phrasing content (fallback).
 - Important attributes: `width` , `height` .
 - Common use cases: Used for drawing graphics via JavaScript (2D and WebGL).
- **<noscript>** : (See 4.1 Document Metadata)

4.12 Web Components

- **<slot>**
 - Categories: Phrasing content (within `<template>` or custom elements).
 - Permitted content: Flow content.
 - Important attributes: `name` .
 - Common use cases: Represents a placeholder in a Web Component's shadow DOM where external DOM content can be inserted.
- **<template>** : (See 4.1 Document Metadata)
- Custom Elements
 - Defined using JavaScript APIs (`customElements.define()`).
 - Can be autonomous (e.g., `<my-component>`) or customized built-ins (e.g., `<button is="my-button">`).
 - The `is` attribute on standard HTML elements is used to associate them with a customized built-in element.

5. Events Reference

This section provides a quick overview of common DOM events.

- Mouse Events: `click` , `dblclick` , `mousedown` , `mouseup` , `mousemove` , `mouseover` , `mouseout` , `mouseenter` , `mouseleave` .
- Keyboard Events: `keydown` , `keyup` , `keypress` (deprecated).
- Form Events: `input` , `change` , `submit` , `reset` , `focus` , `blur` .
- Clipboard Events: `copy` , `cut` , `paste` .
- Drag & Drop Events: `dragstart` , `drag` , `dragend` , `dragenter` , `dragover` , `dragleave` , `drop` .
- UI Events: `resize` , `scroll` , `wheel` .
- Focus Events: `focus` , `blur` , `focusin` , `focusout` .
- Animation Events: `animationstart` , `animationend` , `animationiteration` .
- Transition Events: `transitionstart` , `transitionend` , `transitioncancel` .

Event Bubbling/Capture Summary: Events in the DOM typically go through two phases: 1. Capture Phase: The event travels down the DOM tree from the window to the target element. 2. Bubbling Phase: The event travels back up the DOM tree from the target element to the window. By default, event listeners are triggered during the bubbling phase. The `useCapture` argument in `addEventListener` (or `capture` option in newer APIs) can control whether the listener runs during the capture phase.

6. ARIA & Accessibility Quick Reference

ARIA (Accessible Rich Internet Applications) attributes enhance accessibility for assistive technologies.

- **aria-label** : `<string>` - Provides a text label for an element when visible text is not available or suitable.
- **aria-labelledby** : `<IDREF>` - Refers to the `id` of one or more elements that label the current element.

- **aria-describedby** : <IDREF> - Refers to the `id` of one or more elements that describe the current element.
- **aria-hidden** : `true` | `false` - Indicates that an element and its descendants are not visible to assistive technologies.
- **aria-expanded** : `true` | `false` | `undefined` - Indicates if a controllable element (like a menu) is expanded or collapsed.
- **aria-haspopup** : `true` | `false` | `menu` | `listbox` | `tree` | `grid` | `dialog` - Indicates that an element has a popup associated with it.

Roles: Define the purpose of an element (e.g., `role="button"` , `role="navigation"` , `role="dialog"`). Landmarks: Semantic roles that help users navigate quickly to major sections of a page (e.g., `banner` , `navigation` , `main` , `complementary` , `contentinfo`).

7. Character References

Used to represent characters that are not easily typed or have special meaning in HTML.

• HTML Entities:

- `<` : `<` (less than)
- `>` : `>` (greater than)
- `&` : `&` (ampersand)
- `"` : `"` (double quote)
- `'` : `'` (single quote, not universally supported in older HTML)
- ` ` : Non-breaking space

• Numeric References:

- Decimal: ` ` (non-breaking space)
- Hexadecimal: ` ` (non-breaking space)

8. Common MIME Types

Used in `Content-Type` headers and `<meta>` tags.

- Text:

- `text/html`
- `text/css`
- `text/javascript`
- `application/javascript`
- `application/json`
- `application/xml` , `text/xml`

- Images:

- `image/jpeg`
- `image/png`
- `image/gif`
- `image/svg+xml`
- `image/webp`

- Audio:

- `audio/mpeg`
- `audio/ogg`
- `audio/wav`

- Video:

- `video/mp4`
- `video/ogg`
- `video/webm`

9. Compatibility Notes

Deprecated Elements

These elements are no longer recommended for use. Use their modern alternatives.

- `<big>` : Use CSS `font-size` .
- `<center>` : Use CSS `text-align: center;` .
- `<font>` : Use CSS `font-family` , `font-size` , `color` .
- `<marquee>` : Use CSS animations or JavaScript.
- `<strike>` : Use `<s>` or CSS `text-decoration: line-through;` .
- `<tt>` : Use `<code>` or CSS `font-family: monospace;` .

Obsolete Attributes

Attributes that are no longer part of the HTML standard. Examples include: *

`align` (on many elements): Use CSS `text-align` , `float` , `display` properties. * `b bgcolor` (on `<body>` , `<td>` , etc.): Use CSS `background-color` . * `border` (on `<table>`): Use CSS `border` . * `frameborder` (on `<iframe>`): Use CSS `border` . * `hspace` , `vspace` (on `<img>` , `<object>`): Use CSS `margin` . * `longdesc` (on `<img>`): Use `alt` text or linked content. * `marginheight` , `marginwidth` (on `<iframe>`): Use CSS `padding` . * `scrolling` (on `<iframe>`): Use CSS `overflow` .

10. Minimal HTML Boilerplates

Clean Minimal Document Template

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Document Title</title>
</head>
<body>
  <!-- Content goes here -->
</body>
</html>
```

SEO-Ready Meta Template

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Your Page Title - Engaging Keywords</title>
  <meta name="description" content="A concise and informative description of your page content.">
  <meta name="keywords" content="keyword1, keyword2, keyword3, paragraph1, paragraph2">
  <meta name="author" content="Your Name/Organization">
  <link rel="canonical" href="https://www.yourwebsite.com/your-page">
  <!-- Open Graph / Facebook -->
  <meta property="og:title" content="Your Page Title">
  <meta property="og:description" content="Description for social media sharing.">
  <meta property="og:image" content="https://www.yourwebsite.com/path-to-image.jpg">
  <meta property="og:url" content="https://www.yourwebsite.com/your-page">
  <meta property="og:type" content="website">
  <!-- Twitter -->
  <meta name="twitter:card" content="summary_large_image">
  <meta name="twitter:title" content="Your Page Title">
  <meta name="twitter:description" content="Description for Twitter sharing.">
  <meta name="twitter:image" content="https://www.yourwebsite.com/path-to-image.jpg">
  <link rel="icon" href="/favicon.ico" type="image/x-icon">
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <!-- Content -->
```

```
<script src="script.js" defer></script>
</body>
</html>
```

UTF-8 Safe Setup

The `<meta charset="UTF-8">` declaration in the `<head>` is essential for ensuring that characters from virtually all writing systems are correctly displayed. UTF-8 is the universal standard for encoding text and should be used in all modern HTML documents.

11. Appendix: Full Element Index (Alphabetical)

- `<a>` : Anchor, creates hyperlinks.
- `<abbr>` : Abbreviation or acronym.
- `<address>` : Contact information for author/owner.
- `<area>` : Defines a clickable area in an image map.
- `<article>` : Represents a self-contained content item.
- `<aside>` : Tangentially related content, like a sidebar.
- `<audio>` : Embeds audio content.
- `<b>` : Stylistically bold text.
- `<base>` : Sets base URL for relative URLs.
- `<bdi>` : Bi-directional isolation for text.
- `<bdo>` : Overrides current text direction.
- `<blockquote>` : Long quotation from another source.
- `<body>` : Main content of the document.
- `<br>` : Line break.
- `<button>` : Clickable button.
- `<canvas>` : Area for JavaScript drawing.
- `<caption>` : Caption for a table.
- `<cite>` : Title of a creative work.
- `<code>` : Fragment of computer code.
- `<col>` : Defines column properties for a table.
- `<colgroup>` : Groups columns in a table.

- **<data>** : Links content to a machine-readable translation.
- **<datalist>** : Predefined options for input controls.
- **<dd>** : Description of a term in a description list.
- **** : Deleted text.
- **<details>** : Disclosure widget for revealing/hiding content.
- **<dfn>** : Term being defined.
- **<dialog>** : Dialog box, alert, or other interactive panel.
- **<div>** : Generic container for flow content.
- **document** : Not an element, but the root of the DOM.
- **dl** : Description list.
- **dt** : Term in a description list.
- **** : Emphasized text.
- **<embed>** : Embeds external content.
- **<fieldset>** : Groups related form controls.
- **<figcaption>** : Caption for a **<figure>** .
- **<figure>** : Self-contained content, often with a caption.
- **<footer>** : Footer for its nearest sectioning root.
- **<form>** : HTML form for user input.
- **<h1>** – **<h6>** : Headings of different levels.
- **<head>** : Contains document metadata.
- **<header>** : Introductory content or navigational links.
- **<hgroup>** : Groups a heading and its subheadings.
- **<hr>** : Thematic break between paragraph-level content.
- **<html>** : Root element of an HTML document.
- **<i>** : Stylistically italic text.
- **<iframe>** : Inline frame, embeds another document.
- **** : Embeds an image.
- **"<input>"** : Form control for user input.
- **<ins>** : Inserted text.
- **inert** : Global attribute to make an element non-interactive.
- **<kbd>** : Keyboard input.
- **<label>** : Label for a form control.
- **<legend>** : Caption for a **<fieldset>** .
- **** : List item.

- **<link>** : Links to external resources.
- **<main>** : Dominant content of the document.
- **<map>** : Defines an image-map.
- **<mark>** : Highlighted text.
- **<math>** : Embeds mathematical formulas (MathML).
- **<menu>** : Represents a list of commands or links.
- **<menuitem>** : A command or option in a **<menu>** (legacy).
- **<meta>** : Document metadata.
- **<meter>** : Scalar measurement within a known range.
- **<nav>** : Navigation links.
- **<noscript>** : Fallback content for JavaScript disabled users.
- **<object>** : Embeds external resources.
- **** : Ordered list.
- **<optgroup>** : Groups options in a select list.
- **<option>** : Option in a select list or datalist.
- **<output>** : Result of a calculation.
- **<p>** : Paragraph of text.
- **<param>** : Parameters for an **<object>** .
- **part** : Global attribute for shadow DOM.
- **popover** : Global attribute for popover elements.
- **<pre>** : Preformatted text.
- **<progress>** : Represents progress of a task.
- **<q>** : Short inline quotation.
- **rp** : Fallback for ruby annotations.
- **rt** : Ruby annotation.
- **rtc** : Ruby text container.
- **ruby** : Ruby annotation container.
- **s** : Strikethrough text.
- **<samp>** : Sample output from a program.
- **<script>** : Embeds or references executable code.
- **<section>** : Represents a standalone section of a document.
- **<select>** : Dropdown list.
- **slot** : Placeholder for DOM insertion in Web Components.
- **small** : Side comments or small print.

- **span** : Generic inline container.
- **spellcheck** : Global attribute for spell checking.
- **svg** : Embeds Scalable Vector Graphics.
- **style** : Embeds CSS styles.
- **sub** : Subscript text.
- **summary** : Caption for a `<details>` element.
- **sup** : Superscript text.
- **table** : HTML table.
- **tbody** : Table body section.
- **td** : Table data cell.
- **template** : Client-side content that is not rendered immediately.
- **textarea** : Multi-line text input control.
- **tfoot** : Table footer section.
- **th** : Table header cell.
- **thead** : Table header section.
- **time** : Represents a specific point in time.
- **title** : Document or element title.
- **tr** : Table row.
- **track** : Text track for media elements.
- **translate** : Global attribute for translation control.
- **u** : Underlined text.
- **ul** : Unordered list.
- **var** : Variable in programming or mathematical context.
- **video** : Embeds video content.
- **wbr** : Word break opportunity.

End of document.